Stochastic Contact Graphs: Rethinking the Contact Plan for the 6G Space-Air-Ground Continuum

Supplementary results

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Abstract

This report collects the full studies behind the condensed results of the paper: the evaluation protocol, the orbital skeletons and the fair-design check, the multi-hop continuum in detail, the optimality gap of candidate generation, the comparison with BRUF, RUCoP, and probabilistic CGR, robustness to misspecified, correlated, and learned laws, concurrent flows and finite storage, the placement probe, and the control-plane and computational overhead. Every number and figure here is produced by the companion library `scgraph` (`https://gitlab.inria.fr/jfraire/scgraph`); Section S10 lists the command that regenerates each one. Section, figure, and equation numbers prefixed by S belong to this report; the others refer to the paper.

S1 Evaluation protocol

Held-out scoring. Route selection and evaluation draw from independent random streams, derived from one seed per instance with NumPy's `SeedSequence`. A policy chooses its route with 2000 Monte-Carlo draws per candidate, unless its admission rule asks for more, and the chosen route is then scored on 10 000 fresh draws of the full laws (20 000 in the ε-sweep). Policies that act on what they observe along the way (hop-by-hop repair, ranked tables, BRUF and RUCoP, the placement probe) are scored on fresh realisations of the whole graph, shared by every policy they are compared with, so that each comparison is paired. No number in the paper or here rests on the draws that selected the route it describes. In the submitted version the same draws served both purposes; scoring held out changed the reported numbers by at most a few tenths of a point, except that routes which showed zero in-sample violations now show their small true rate (for instance 1.3×10^{-4} and 5×10^{-5} on the mobile event at $\sigma = 15$ and 20 s).

Intervals. An interval on a single proportion is the exact two-sided Clopper-Pearson interval; error bars on means across instances are 95% intervals. A measured zero over M draws is reported as its exact one-sided 95% upper bound, $1 - 0.05^{1/M} \approx 3/M$.

Admission. Lower-bound admission is Eq. (12) of the paper, with the confidence level split over the candidate set (Bonferroni). The number of selection draws follows Eq. (13) of the paper with a factor of three of headroom and a floor of 2000, which gives 16 674 draws per candidate at $\varepsilon = 10^{-3}$, $\delta = 0.05$, and a typical candidate set. When no candidate is admitted, the router returns the candidate with the highest estimated delivery probability as a flagged best-effort route; the evaluations count such requests as infeasible and score the flagged route as executed.

S2 Orbital skeletons and the fair-design check

The two-hop skeleton. CPD propagates a Walker-Delta 12/3/1 shell (twelve satellites in three planes at 600 km and 53°) over six hours, with an event site and a gateway 390 km apart in western Europe and a 10° elevation mask, computes the accesses, resolves interface conflicts by fair contact-plan design, and exports the plan (Fig. S1). The event site has a usable satellite in view for about a fifth of the horizon, spread over 35 separate windows. At a 10° mask a satellite at 600 km covers a radius of roughly 1760 km, so both sites are usually in view at once, but fair design gives each satellite one interface at a time and serves the two sites in consecutive, non-overlapping slots: not one of the 35 uplink windows overlaps a downlink window on the same satellite. The bundle is therefore held on board across the interface switch, and end-to-end latency, set by pass geometry, ranges over hundreds of seconds. Contact durations span roughly 30 to 390 s depending on how near overhead a pass goes.

Fair design is not what produces the result. Fair design is a property of the plan, so we re-ran the mobile event on the same constellation without the design stage (`cpd compute`), where every pass serves both sites at once and all 15

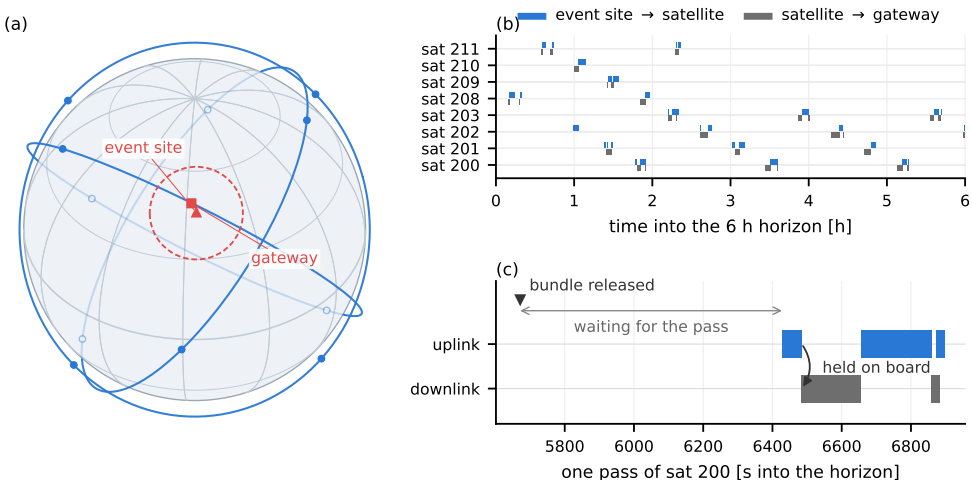


Figure S1: The two-hop orbital skeleton. (a) The constellation handed to CPD at epoch, with the event site, the gateway 390 km away, and the event site's 10° visibility cap (dashed). (b) Every ground window over the six-hour horizon, one row per satellite, omitting the four satellites that never see either site. (c) One pass magnified, showing the consecutive, non-overlapping uplink and downlink slots that make a bundle wait on board between them.

uplink windows overlap a downlink window on the same satellite (Fig. S2). Deterministic CGR then violates 28.8% of deadlines at $\sigma = 60$ s against zero for the SLA-aware router, whose mean delivery time is 583 s. The delay is set by the wait for the next pass, which the release-time band of the paper fixes at 200 to 900 s in both plans.

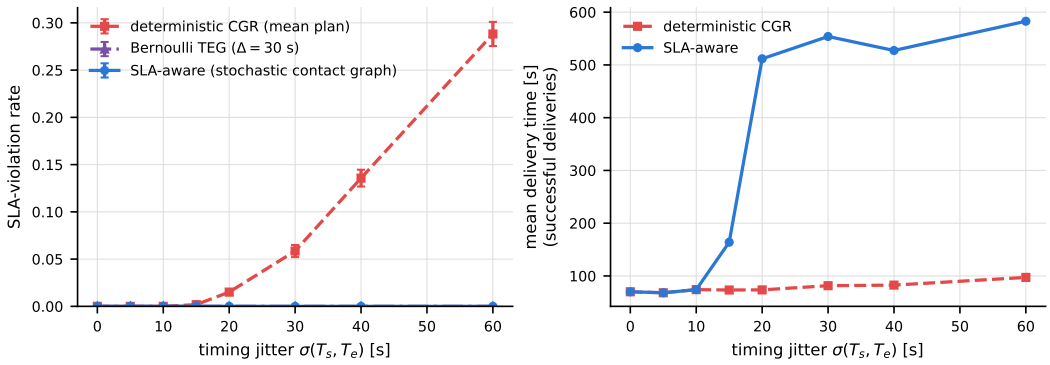


Figure S2: The mobile event on the plan without fair contact-plan design: SLA-violation rate (left) and mean delivery time of delivered bundles (right) against timing jitter σ. Same protocol as Fig. 5 of the paper.

The multi-hop skeleton. The multi-hop continuum runs on a second CPD plan: a Walker-Delta 60/6/1 shell at 550 km and 53°, ten satellites per plane, with an in-plane link to each neighbour and cross-plane links to the adjacent plane within 5000 km, a gateway in Paris, three terminals at increasing distance from it (Barcelona, Rabat, and Dakar, at 830, 1800, and 4200 km), and a stratospheric platform at 20 km beside each terminal. Satellites have six interfaces and the gateway two, so passes overlap. The six-hour plan holds 3606 contacts, and the base experiments use the Dakar terminal; the generality study adds 40- and 90-satellite variants (the smallest without cross-plane links) and the nearer terminals. Table S1 lists the laws attached to each link class. Each instance draws a release time uniformly over the plan, restricts the graph to the contacts whose support overlaps the following 1800 s, injects three drones with randomised nominal windows, and sets the deadline a margin after the nominal arrival of the best route that avoids the drones (the reliable path), 60 s by default.

S3 The multi-hop continuum in full

Fig. 9 of the paper reports thirty instances per point; committed routes are scored on 10 000 held-out draws each, and repaired execution on 1000 held-out realisations of the whole graph shared by all policies.

Committed routes. CGR on the nominal plan violates the deadline 18% of the time even without drone jitter, because a drone may be absent and acquisition delays and fades cut windows on its tight nominal connections, and

its violation climbs to 37% at $\sigma = 40$ s and 47% at $\sigma = 60$ s. The SLA-aware router holds 8% to 10% across the sweep, choosing per instance between the reliable narrowband path and drone routes with slack. The Bernoulli corner fails 60% to 84% of the time at a 10 s grid, because one hop per slot cannot fit routes of seven to nine hops into the deadline; this is the resolution loss of the corner ablation at a depth where it binds.

Repaired execution. Committed routes understate what a deployed CGR does, since a node that finds its contact missing re-routes. Under hop-by-hop repair, CGR becomes a strong baseline, with 0.8% violation without jitter and 12.5% at $\sigma = 40$ s. The SLA-aware planner that scores its first route under the same repair rule keeps an edge at every jitter level, with 7.7% at $\sigma = 40$ s and 9.0% against 22.7% at $\sigma = 60$ s. The deadline sweep shows the same order at every margin, from 30.4% against 22.1% sixty seconds below the reliable path's arrival to 8.9% against 0.1% at 240 s above it, where the SLA-aware planner takes the reliable path and CGR, ranking by nominal arrival, keeps starting on a drone.

Compounding. On a controlled chain of k jittered hops with fixed slack (Fig. S3), the true probability of missing the deadline at $\sigma = 30$ s rises from 21% at one hop to 76% at ten, while the product of per-hop success probabilities, which treats hops as independent and lateness as non-propagating, stays between 21% and 35%. At small jitter the error flips sign: the slack absorbs lateness, the true 95% lateness stays at 17 s at every depth, and the sum of per-hop quantiles grows to 164 s at ten hops. Per-hop reasoning therefore errs in both directions as routes deepen, while the end-to-end recursion of Section V-A of the paper, evaluated by Monte Carlo, is exact at any depth.

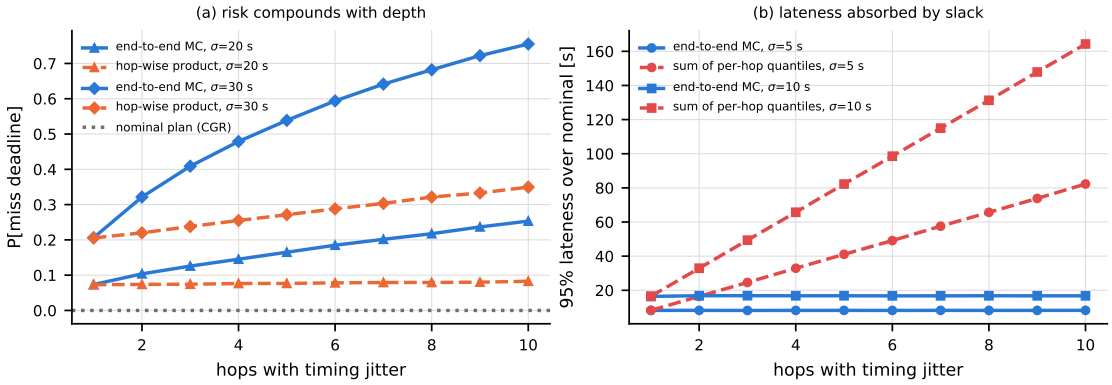


Figure S3: How timing jitter compounds along a route of k jittered hops with 10 s of nominal slack between them. (a) Probability of missing the deadline (nominal arrival plus 30 s): exact end-to-end Monte Carlo against the product of per-hop success probabilities. (b) The 95% lateness over the nominal arrival: exact end-to-end against the sum of per-hop 95% quantiles. 2×10^5 draws per point.

The joint law. A planner given only the marginals of the drone windows, without their coupling, chooses routes that perform the same as the joint planner's (8.4% against 8.3% held-out violation at $\sigma = 40$ s). Where the deadline forces a drone hand-off (zero margin), however, the delivery probability it states for its route is off by 7 points, while the joint planner's statement is within 0.2 points of the held-out truth. In this use case the joint law matters less for which route is chosen than for the probability attached to it, which is the quantity SLA admission rests on.

Generality. Around the base point we varied one factor at a time over 25 instances per level (Fig. S4): the terminal's distance to the gateway (830 to 4200 km, routes of four to seven hops), the constellation (40, 60, and 90 satellites), the deadline margin, the jitter, the bundle size, and the family of the timing law (normal, Student-t with three degrees of freedom, uniform, all with the same standard deviation). Within each execution model the SLA-aware policy beats its CGR counterpart at all thirteen points, with paired 95% intervals that exclude zero: by 0.12 to 0.39 with committed routes and by 0.004 to 0.125 with repair. Distance, density, and timing-law family barely move its violation rate, which stays between 8.5% and 9.4% with committed routes. The gain shrinks at a zero margin, where the reliable path is never fast enough, at small jitter, where repair alone delivers 98%, and for a 200-unit bundle, which makes the reliable path itself fragile.

S4 How much the candidate set misses

Because quantiles do not compose, the router ranks an enumerated candidate set instead of searching the route space (Section V-C of the paper). For each instance we enumerated every node-simple route whose *support envelope* arrives by the deadline. The envelope replaces each contact by its most favourable corner: earliest start, latest end, highest rate, and existence whenever some latent state allows it. Because the arrival recursion is monotone in each of these, a route that some realisation delivers on time also meets the deadline under the envelope, so the enumerated set provably contains every route that can ever deliver on time (up to the hop bound and route cap, neither of which was hit, at

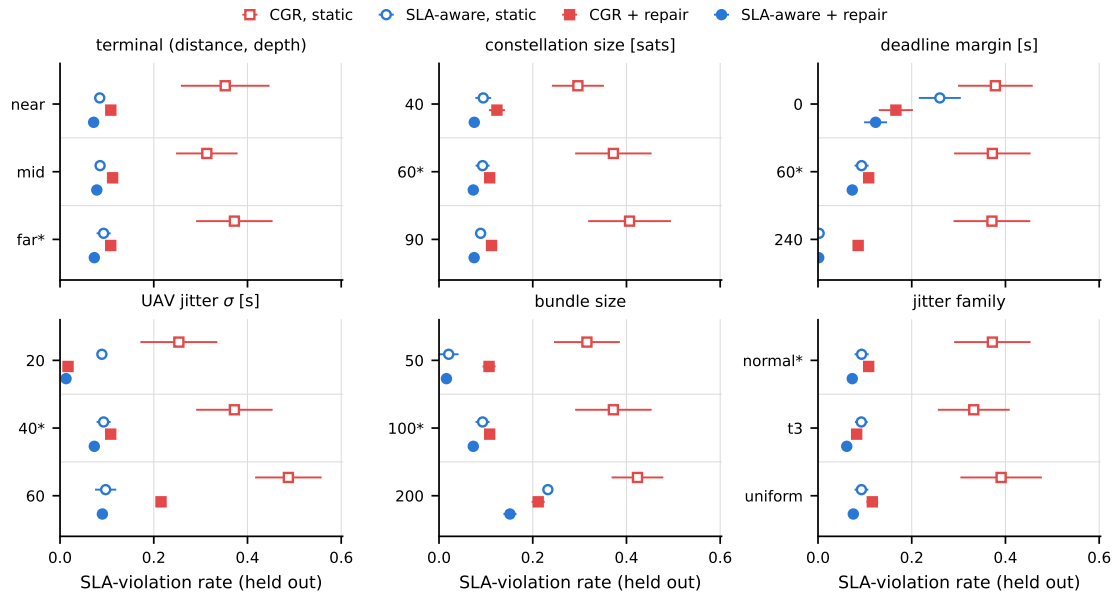


Figure S4: One-at-a-time sensitivity of the multi-hop continuum around the base point (starred: far terminal, 60 satellites, 60 s margin, σ = 40 s, 100-unit bundle, normal jitter). Held-out SLA-violation rate of CGR and the SLA-aware policy, committed (open markers) and with hop-by-hop repair (filled). Twenty-five instances per level, 95% intervals.

up to 188 000 routes per instance). All enumerated routes were scored on shared draws, and the router's choice was compared with the best of them on 20 000 fresh paired draws (Table S2, Fig. S5).

On the mobile event the best route was in the candidate set in all 160 instances, and the router chose it, or a route statistically tied with it, in 96% to 100% of instances at the default margin. Its misses cost at most 0.002 of delivery probability and are ranking noise from the 2000 selection draws. The multi-hop continuum is where a candidate generator can fail, and the default one did. Without the risk-class routes, the best route was offered in only 25% of instances at a 60 s margin, with a mean loss of 0.081. The safe route rides the slow narrowband link and shares its uplink and feeder with the fast drone routes, so neither a realisation's earliest-arrival route nor contact suppression reaches it. Adding the risk-class routes (mean-plan routes with a whole class of uncertain contacts excluded, Section V-C of the paper) raises the share to 95% and cuts the mean loss to 0.0004 for about 5% more selection time, and all multi-hop results use them. At a zero margin, where every good route needs a drone hand-off, the residual loss averages 0.007. Infeasibility is mostly intrinsic. Of the 345 requests the router declared infeasible under lower-bound admission, 305 had no enumerated route meeting the budget either, and no admitted route exceeded its budget on held-out draws.

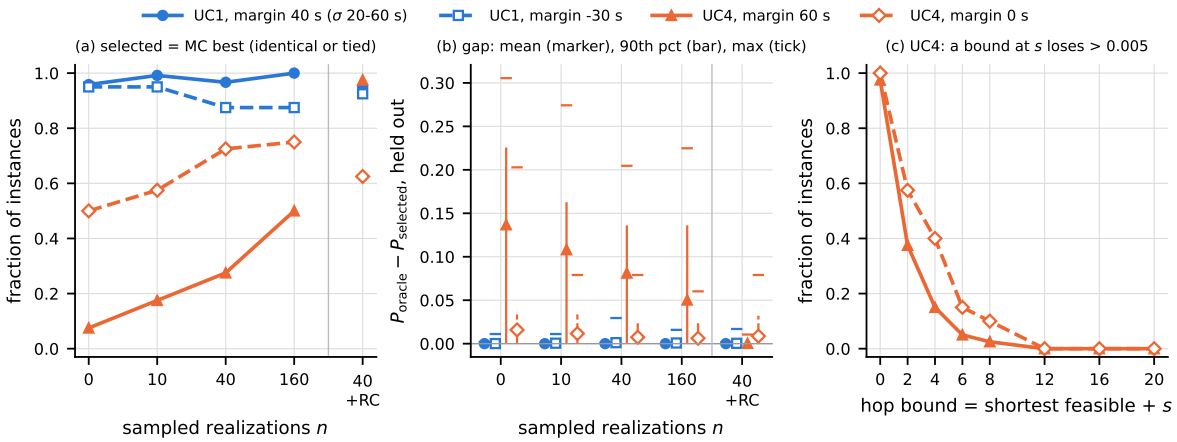


Figure S5: Optimality gap of candidate generation. (a) Fraction of instances in which the selected route is the Monte-Carlo best (identical or statistically tied), against the number n of sampled realisations used to generate candidates, and with the risk-class routes added (+RC). (b) Held-out loss against the best enumerated route: mean (marker), 90th percentile (bar), maximum (tick). (c) Multi-hop continuum: fraction of instances in which bounding the hop count at the shortest feasible count plus s would lose more than 0.005.

S5 Against BRUF, RUCoP, and probabilistic CGR

We reimplemented BRUF and RUCoP [1, 2] by exact backward induction over (copy distribution, slot) states, with acknowledged communication (a failed transmission leaves the copies at the sender) and the reachability pruning of the original tooling, and validated both against the reference toy network. They run on the Bernoulli projection (Eq. (17) of the paper) of the same instances. Their output is a contingent policy: a copy whose contact fails stays at its node, and the policy acts again in the next slot. We execute that policy against the same held-out realisations of the full laws as our routers, letting a copy leave as soon as its planned contact opens, which is how static routes are executed as well, so that static and contingent policies differ only in contingency; a slot-locked variant changes the conclusions only at coarse grids. A simplified probabilistic CGR in the style of [3] ranks candidates by the product of the contacts' marginal existence probabilities and breaks ties by nominal arrival.

Timing uncertainty (Fig. S6). Contingency buys nothing here. On the mobile event at $\sigma = 60$ s, BRUF violates the deadline 9.2%, 3.3%, and 0.55% of the time at slots of 60, 30, and 10 s, against 29.2% for CGR and no violation in 80 000 runs for the SLA-aware router (95% upper bound 4.6×10^{-5}). BRUF matches the static Bernoulli path at every slot width, because its failures occur at relays with no fallback contact, so its loss is the resolution loss of the corner ablation. Probabilistic CGR coincides with deterministic CGR, since every contact of this case exists.

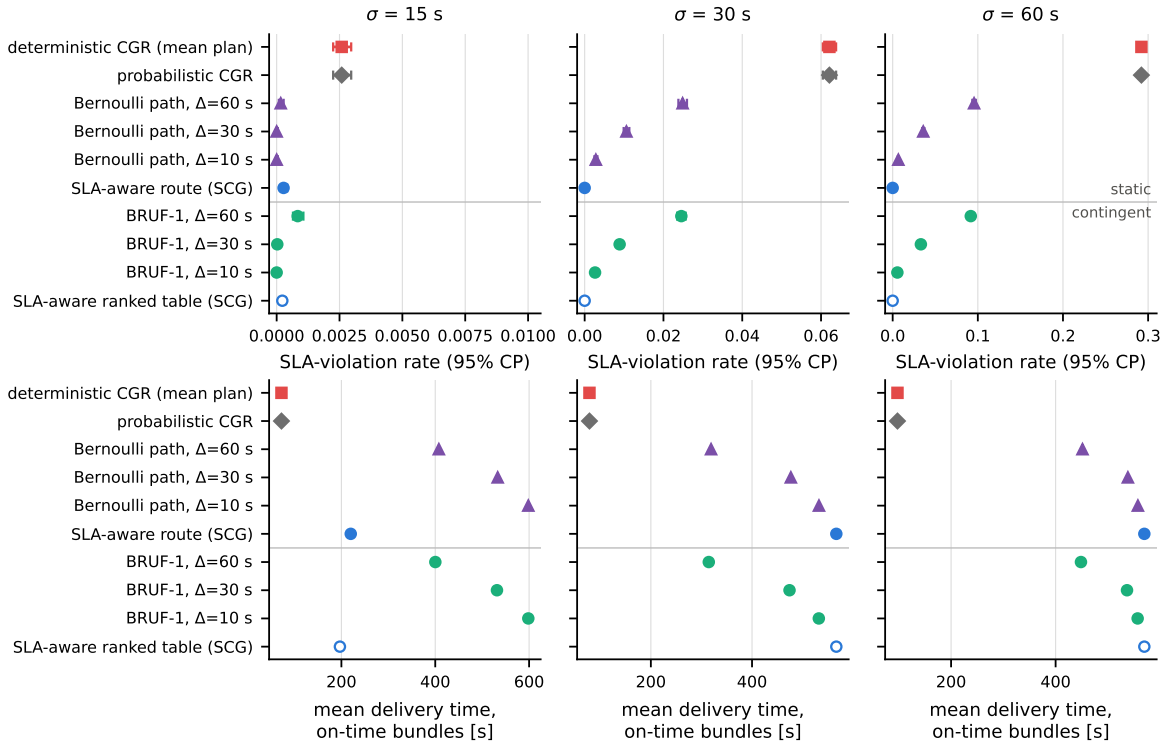


Figure S6: The mobile event at $\sigma = 15$, 30, and 60 s: SLA-violation rate (top, exact 95% intervals) and mean delivery time of on-time bundles (bottom) for static policies (CGR, probabilistic CGR, the Bernoulli path at three slot widths, the SLA-aware route) and contingent ones (BRUF at three slot widths, the SLA-aware ranked table).

Correlated existence failures (Fig. S7). Contingency pays, and RUCoP outperforms every committed replica set. At $\rho = 0.5$ with two copies, RUCoP delivers 73.3% against 65.0% for our correlation-aware committed set, 60.2% for the Bernoulli selection, and 48.9% for naive replication and for probabilistic CGR, while an oracle that sees the realisation reaches 75.8%. The gain comes from acknowledged retries after an observed failure, and telling RUCoP the latent state adds only 0.8 points. Its own model, however, cannot say how well it will do. RUCoP predicts 98.2% delivery where it achieves 73.3%, and refining its grid to 10 s raises the prediction to 100% while the achieved ratio falls to 70.7%. The committed sets of the stochastic contact graph achieve less and state what they achieve. A contingent policy computed over the full laws would do both, and it is the natural next step.

Tractability (Fig. S8). Exact BRUF and RUCoP are cheap on these small cases, with RUCoP-3 solved at one-second slots in under half a second. Their state space binds with copies on large plans: on a two-hour, 48-satellite plan, RUCoP-3 needs 5.1 million states at ten-second slots, and RUCoP-4 exceeds six million already at sixty-second slots, where route selection on the stochastic contact graph takes 0.33 s.

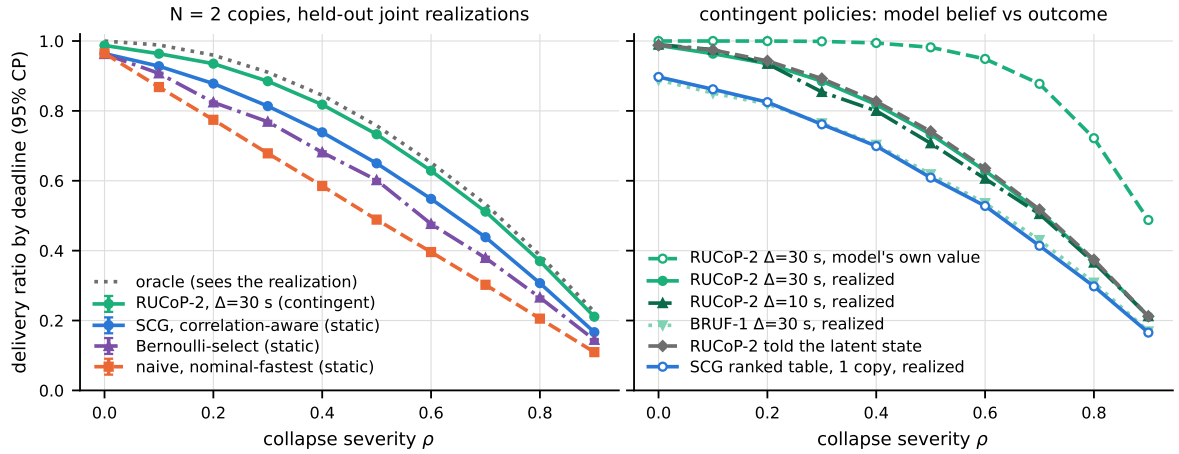


Figure S7: RUCoP against committed replica sets on the emergency case ($N = 2$ copies). Left, delivery ratio by the deadline on held-out joint realisations. Right, for the contingent policies, the delivery each model predicts for itself against what it achieves. Forty instances per point, exact 95% intervals.

S6 Misspecified, correlated, and learned laws

SLA admission is a statement relative to the laws, so we measured what happens when the planner's laws are wrong.

Heavy tails (Fig. S9). On the mobile event at $\sigma = 20$ s, where the router keeps air relays, we let the true start-time law be a Student-t with the same standard deviation as the planner's Gaussian, from $\nu = 10$ down to 2.5 degrees of freedom. With a 5% budget the chance-constrained router holds: its held-out violation grows from 1.0% to 2.0% and no instance exceeds the budget. With a 1% budget it does not. The mean stays below 1%, but 9%, 47%, and 63% of instances exceed the budget at $\nu = 5$, 3, and 2.5, and at $\nu = 2.5$ the planner states 0.18% where the truth is 0.89%. Planning on a t_3 belief restores the 1% budget for every tail, at a mean delivery time of 565 s against 231 s, because the heavier belief retreats to the backbone. Tight budgets thus demand the right tail family, and a planner that does not know it can buy safety with a heavier-tailed belief.

Capacity: misspecified spread and correlated passes (Fig. S10). The capacity saving rests on two properties of the laws, their spread and their independence across passes. When the planner underestimates the spread, the chance-constrained plan still meets the SLA at the two moderate severities (0.3 and 0.5) with a true coefficient of variation 1.5 times the planned one, and fails at twice it, delivering 91.9%. Correlation between passes is the sharper threat. On the pass sequence of the paper the plan tolerates $\varrho = 0.2$, but across all 27 possible starting passes a correlation of 0.1 already breaks it in 11% to 63% of starts, depending on severity. A planner that models the correlation through the common factor restores the SLA, and its saving over worst-case sizing then shrinks from 2.5× to 2.0× at $\varrho = 0.2$ and 1.5× at $\varrho = 0.5$, vanishing at $\varrho = 1$, where the passes fade together and their sum no longer concentrates. A planner that assumes $\varrho = 0.3$ when the passes are in fact independent reserves 16% more capacity, meets the SLA from every starting pass up to that correlation, and still sits 26% below worst-case sizing.

Strength and structure of latent dependence (Fig. S11). We coupled the districts' collapses in the emergency case: with probability λ the coupled districts share one collapse draw, otherwise each draws its own, so every marginal is unchanged. When all districts are coupled, the gain of correlation-aware replication over correlation-blind replication falls linearly from 0.16 to zero as λ goes to one, since no placement escapes a collapse that takes every district down; a planner that assumes independent districts picks the same copies but overstates its own delivery by 0.16λ. When only two of three or four districts are coupled, the correlation-aware planner holds 0.65 at every λ by never placing both copies in the coupled pair, while the planner that assumes independence loses up to 0.06.

Learning the law (Fig. S12). On the mobile event at $\sigma = 40$ s, the planner observes n windows of the true jitter, fits σ, and routes on its fit with lower-bound admission at a 5% budget. A plug-in fit underestimates σ often enough to breach the budget in 42% of instances after three windows, 20% after ten, 3.7% after fifty, and none after two hundred. Planning on a 90% chi-square upper bound for σ keeps breaches at or below 4% at every n, with a delivery time within 19 s of an oracle that knows σ. At $\sigma = 20$ s the same planner pays more while data is scarce, 429 s against 77 s after three windows, and comes within 10 s of the oracle by fifty. Parameter uncertainty is therefore one more spread the planner can carry, like the staleness of Section S9.

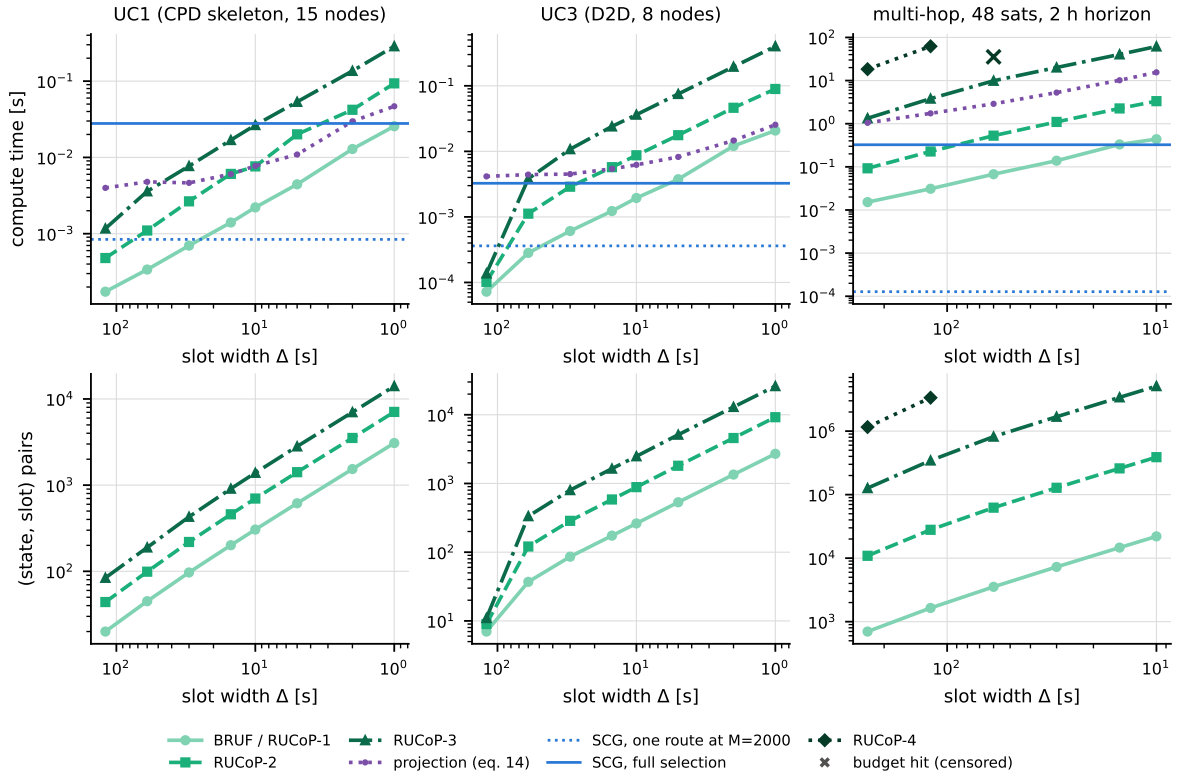


Figure S8: Cost of exact BRUF and RUCoP-N against the slot width on the mobile event, the emergency case, and a two-hour, 48-satellite multi-hop plan: compute time (top) and (state, slot) pairs (bottom), with the SLA-aware selection for reference (one route at $M = 2000$, and a full selection). Crosses mark runs that hit the computation budget.

S7 Many bundles: contention, residual volume, and storage

The model routes one atomic bundle with unbounded storage, and with competing bundles a new risk appears: if every bundle that misses an aerial window falls back on the same pass, that pass and the satellite's storage become the bottleneck. We released bursts of K bundles at the event site ($\sigma = 40$ s, 30 instances) and executed them on held-out realisations with an event-driven simulator in which a contact serves one bundle at a time in arrival order, an atomic transfer cut by the window end is lost, and relay storage holds at most B bundles (Fig. S13). Four policies route the burst: CGR on the nominal plan, CGR with SABR-style booking of residual contact volume, the single-bundle SLA-aware router applied to each bundle, and a contention-aware SLA-aware router that assigns bundles in turn, scoring each candidate by Monte Carlo with the bundles already assigned.

The single-bundle router congests as one would fear. It sends the whole burst to the one favourable pass, which carries about four bundles before the deadline, so its on-time fraction falls from 100% at $K = 1$ to 52.5% at $K = 8$ and 26.3% at $K = 16$, no better than plain CGR, while the satellite needs storage for about ten bundles. Contention-aware scoring removes most of that loss, with 98.9%, 93.8%, and 84.8% of bundles on time at $K = 4$, 8, and 16, against 81.7%, 79.0%, and 74.4% for deadline-blind CGR with volume booking, at a mean delivery time of 284 s against 128 s. With storage for two bundles per relay at $K = 16$, it delivers 59% against 51% for booking and 17% for CGR. Splitting each bundle into two fragments lowers the on-time fraction from $K = 4$ on (84.8% to 78.0% at $K = 16$), because a bundle now needs both halves on time. The stochastic contact graph thus extends to flows by scoring candidates against the traffic already committed, and the single-bundle analysis of the use cases is the $K = 1$ corner of that extension.

S8 The placement probe

The three fixed placements of Section VI-B of the paper are executed against the same drawn realisations, so the observed differences are attributable only to the combination of planning location, global knowledge, and online adaptation.

Two-hop use cases (Fig. S14). In the emergency case, the ground-central placement is uniformly the worst: at $\rho = 0.4$ it delivers 0.49 against 0.67 for the sky proxy and 0.71 for the distributed policy, a spread of 0.22 in delivery

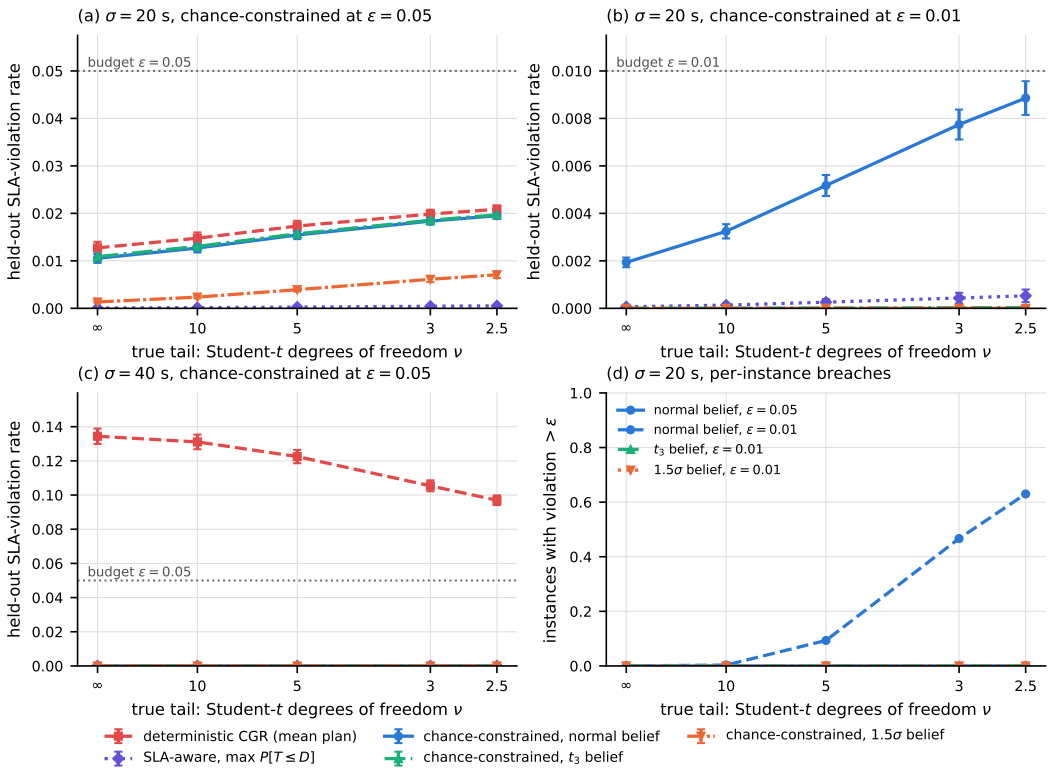


Figure S9: Heavy-tailed true timing on the mobile event, against the degrees of freedom ν of the true Student-t start law. (a, b) Held-out violation at $\sigma = 20$ s under chance-constrained routing at $\varepsilon = 0.05$ and $\varepsilon = 0.01$, for a normal, a t_3, and a 1.5σ belief; (c) the same at $\sigma = 40$ s and $\varepsilon = 0.05$; (d) fraction of instances whose violation exceeds the budget.

ratio, or a factor of 1.45. A single uplinked route has no answer when one of its contacts fails to materialise, and under a shared collapse that happens often. In the mobile event, the picture inverts with the regime. While the deadline leaves room for the timing-certain backbone, the two global planners find it and deliver everything, and the distributed policy pays about 2.9 points for its lack of a downstream view, because ranking its own outgoing contacts by earliest arrival walks it into the jittered air relays. Once the deadline drops below the backbone's arrival, the global planners fall to 0.90 while the distributed policy holds 0.97.

The multi-hop continuum (Table S3, Fig. S15). Two-hop topologies cap how much a local-view penalty can express, so we repeated the probe on the multi-hop continuum. There the distributed placement delivers nothing by the deadline at any margin: with no downstream view it bounces along short inter-satellite links, and only 1.5% of bundles reach the gateway at all within 200 hops. The sky proxy, whose ranked tables are keyed by node and by four coarse time buckets, falls below the single ground route at every margin, because such tables send bundles round loops on a deep route. The ground route is the best of the three placements throughout. A related inversion survives: reactive CGR, which re-plans locally over the full nominal plan, beats the single ground route at tight deadlines (0.83 against 0.75 at a zero margin) and loses to it with slack (0.89 against 0.97 at 120 s). The SLA-aware plan with local repair is best at every margin. On a deep route a planner without a downstream view fails outright, which confirms that the two-hop setting flattered the distributed policy, and the strongest combination is a global plan computed under the laws and repaired locally when a contact fails.

S9 Control-plane and computational overhead

Sidecar size. A contact whose laws are all point masses carries no annotation, so the sidecar of the mobile event, whose satellite contacts are deterministic, takes 126 bytes at any plan size. In the worst case, with a full joint law on every contact (start and duration laws, a fade law, an existence mass, group and factor identifiers, and three copula coefficients), the sidecar is 1.5 times the size of the ION text plan it annotates: 729 KiB for the 14 640-contact plan, or six seconds of uplink at 1 Mbit/s. Refreshing every air and ground contact of the multi-hop use case takes 8.2 s at 100 kbit/s.

Route selection against SABR (Fig. 10 of the paper). A single CGR search is the wrong reference, because SABR does not stop at one route. The route table it builds anyway, with its alternative routes, costs 0.11 to 385 ms on the

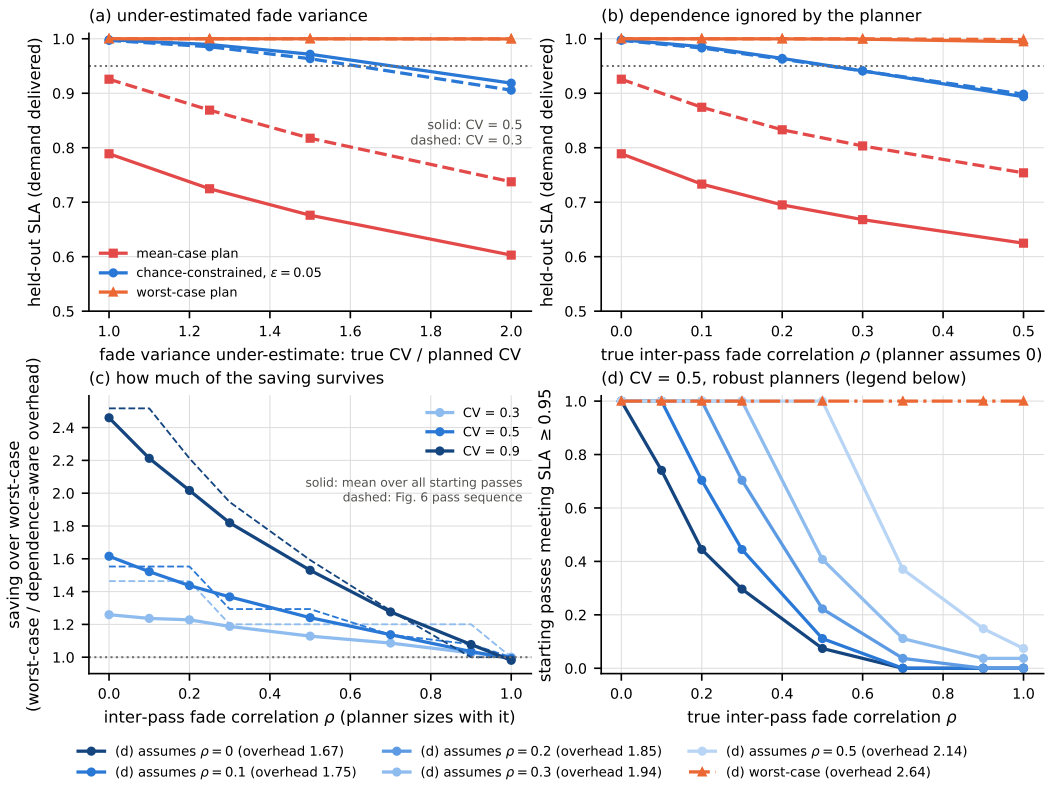


Figure S10: Logistics under misspecified laws (ρ in the figure is the inter-pass correlation ϱ of the paper). (a) Held-out SLA against the ratio of true to planned coefficient of variation; (b) against an inter-pass correlation the planner ignores; (c) saving over worst-case sizing that survives when the planner sizes with the correlation; (d) share of starting passes meeting the SLA, at CV 0.5, for planners that assume a given correlation.

plans of 98 to 14 640 contacts, so a complete SLA-aware selection costs 288 times more on the smallest plan, 12 times at 1110 contacts, and 3.4 times at 14 640. Most of the remaining gap on large plans comes from re-sampling the fixed-time satellite contacts in every realisation, and copying them instead brings selection to 0.49 s, 1.27 times the SABR table. None of this runs onboard: the forwarding plane resolves a precomputed table entry in 0.53 to 0.60 μs per decision, flat in plan size and about a thousand times cheaper than one CGR search on the largest plan, from a table of 138 to 486 bytes per flow. These timings come from a single-threaded Python implementation on a laptop processor, measured on an idle machine; a flight implementation would change the constants and keep the shape of the comparison.

Staleness (Fig. S16). We let the drone windows drift as a random walk of rate $\kappa = 1\,\mathrm{s}/\sqrt{\mathrm{s}}$ between the moment the planner's information was current and the moment the bundle flies, so information of age Δ_a carries an extra spread $\kappa\sqrt{\Delta_a}$. At $\sigma = 40$ s every SLA-aware variant stays on the timing-certain backbone and records no violation up to an hour of information age, while CGR degrades from 14% to 36%. At $\sigma = 10$ s, where the routers use the air relays, a planner that ignores the age of its data degrades like CGR, to 26% after an hour, while one that widens its law to $\sigma_{\mathrm{eff}} = \sqrt{\sigma^2 + \kappa^2\Delta_a}$ stays at or below 1.5%. It pays by retreating to the backbone, with mean delivery times up to 554 s against 107 s for a planner with fresh information.

S10 Reproducing the results

Every study is one `make` target in the `scgraph` repository (Table S4). Each target writes its numbers to `results/` and its figure to `figures/`. The target `reproduce` regenerates the submitted studies, and `reproduce-revision` the new ones. Runs are seeded and parallelised over instances with independent `SeedSequence` streams, so results do not depend on the number of workers. Longer notes on each study, with every intermediate number and design choice, are in `results/notes/`.

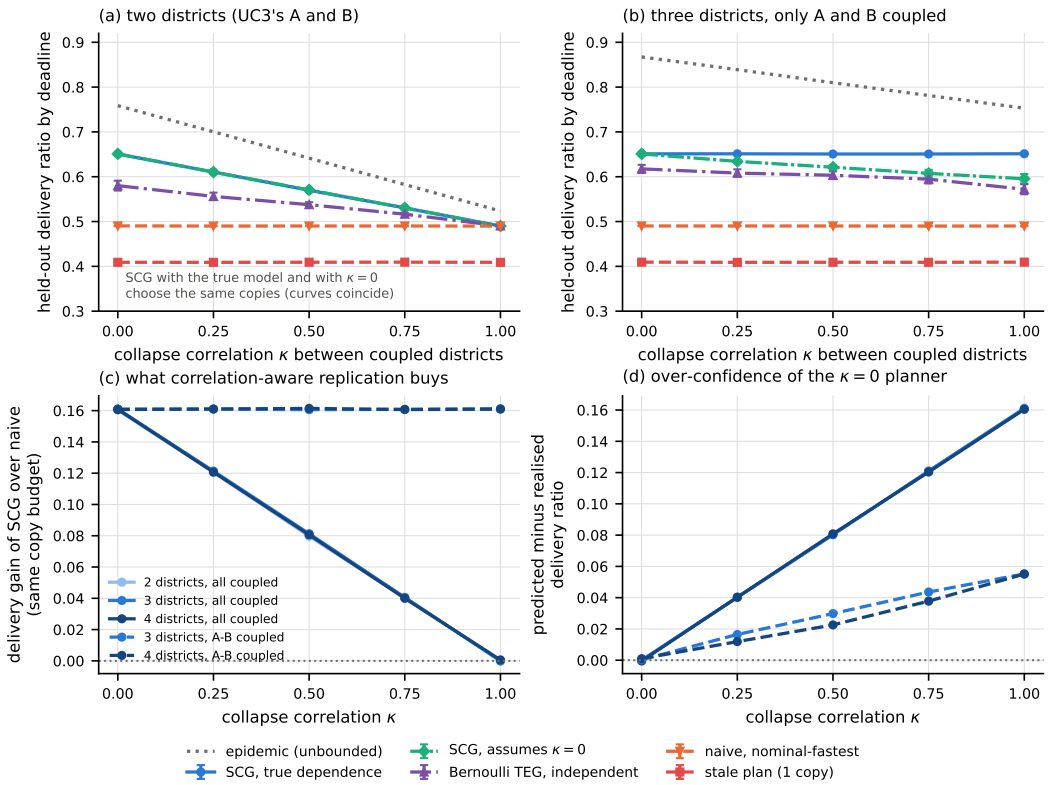


Figure S11: Strength and structure of the dependence in the emergency case (κ in the figure is the coupling λ of the text). (a) Two coupled districts; (b) three districts with only A and B coupled; (c) delivery gain of correlation-aware over naive replication at the same copy budget; (d) over-confidence of the planner that assumes independent districts.

References

[1] F. D. Raverta, R. Demasi, P. G. Madoery, J. A. Fraire, J. M. Finochietto, and P. R. D'Argenio, "A Markov decision process for routing in space DTNs with uncertain contact plans," in *Proc. IEEE Int. Conf. on Wireless for Space and Extreme Environments (WiSEE)*, 2018, pp. 189–194.

[2] F. D. Raverta, J. A. Fraire, P. G. Madoery, R. A. Demasi, J. M. Finochietto, and P. R. D'Argenio, "Routing in delay-tolerant networks under uncertain contact plans," *Ad Hoc Networks*, vol. 123, p. 102663, 2021.

[3] F. Walter and M. Feldmann, "Leveraging probabilistic contacts in contact graph routing," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, Waikoloa, HI, USA, 2019.

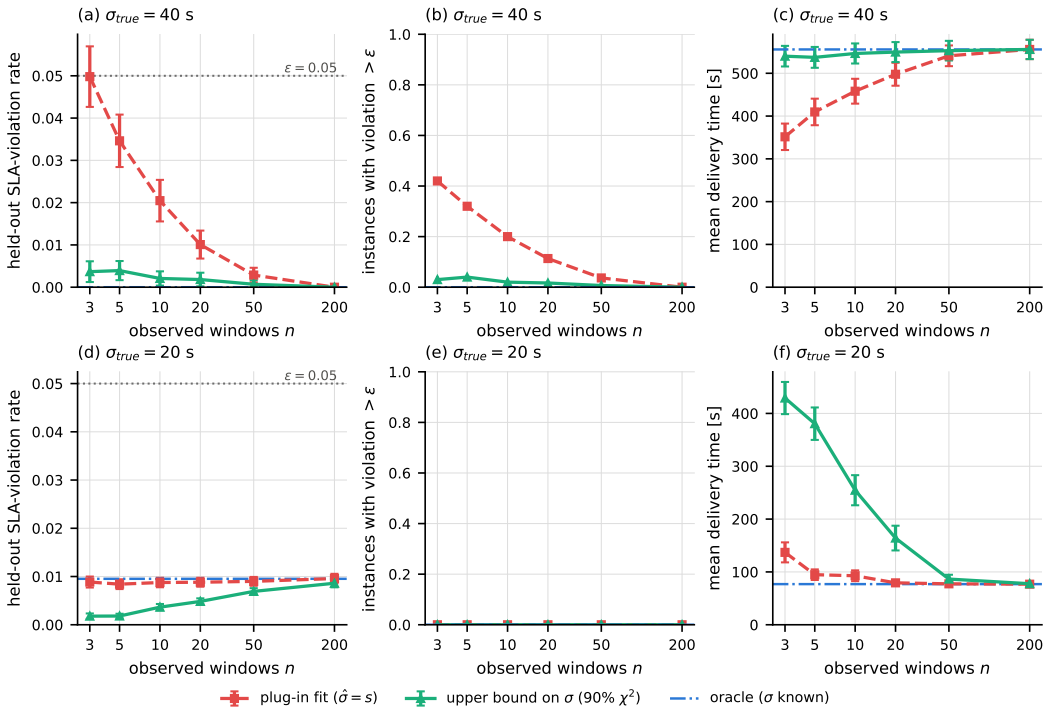


Figure S12: Learning the timing law from n observed windows on the mobile event: a plug-in fit, a 90% chi-square upper bound on σ, and an oracle that knows σ, at $\sigma_{\text{true}} = 40$ s (a–c) and 20 s (d–f). Held-out violation, fraction of instances breaching $\varepsilon = 0.05$, and mean delivery time.

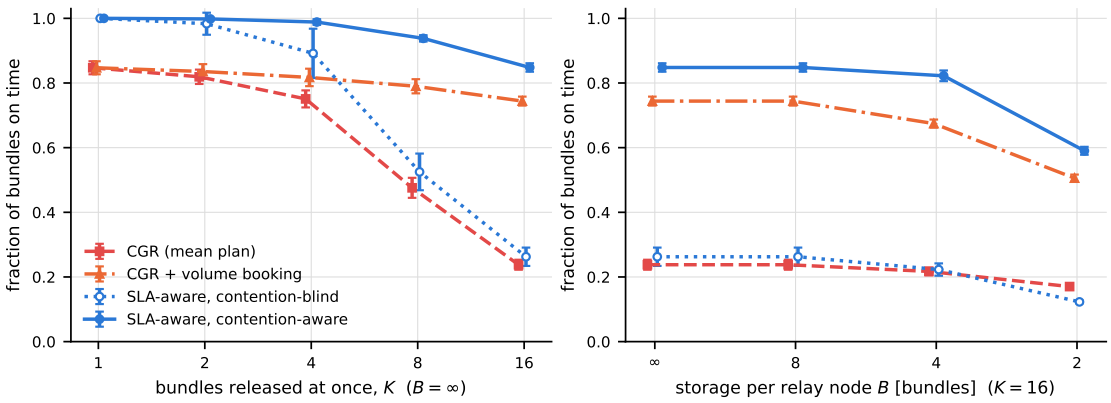


Figure S13: Bursts of K bundles at the event site ($\sigma = 40$ s). Left, fraction of bundles on time against K with unbounded storage; right, against the storage B per relay node at $K = 16$. Thirty instances, 1000 held-out realisations each; 95% intervals across instances.

Table S1: Laws of the multi-hop continuum, per link class. Rates in volume units per second; the bundle is 100 units.

Link class	Timing	Existence	Rate
terminal → HAPS, narrowband	certain, open over the whole 1800 s window	1	0.5
terminal → drone, drone → HAPS	start truncated normal about nominal, s.d. σ; duration 50–80 s; Gaussian copula with start-duration −0.8, start-rate −0.5, duration-rate 0.4	0.9 (first hop)	lognormal, mean 10, CV 0.3
HAPS → LEO	certain	1	lognormal, mean 10, CV 0.2
in-plane ISL	certain	1	50
cross-plane ISL	start late by an acquisition time, truncated normal (mean 20 s, s.d. 10 s); end fixed (start-duration correlation −0.99)	1	50
LEO → gateway feeder	certain	1	lognormal, mean 2, CV 0.5; common weather factor, $\varrho = 0.3$
terminal → LEO, direct-to-satellite	certain	0.35	2

Table S2: Candidate set against an exhaustive enumeration of every route that can meet the deadline. Loss: best enumerated delivery probability minus that of the chosen route, on 20 000 fresh paired draws.

Setting	best offered	chosen or tied	mean loss	max loss
Mobile event, margin 40 s	100%	96–100%	≈0	0.002
Mobile event, margin −30 s	100%	88–95%	0.001	0.030
Continuum, 60 s, default set	25%	28%	0.081	0.205
Continuum, 60 s, with risk-class routes	95%	97%	0.0004	0.011
Continuum, 0 s, default set	78%	72%	0.007	0.079

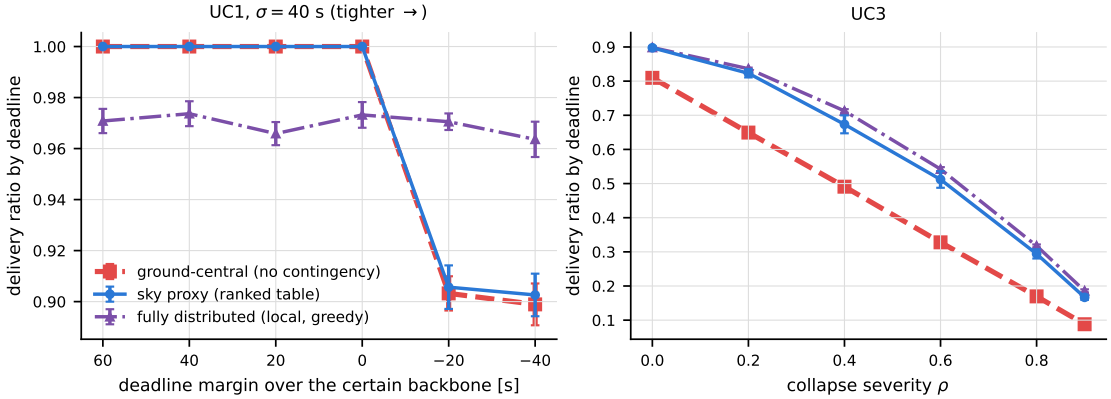


Figure S14: Placement probe on the two-hop use cases: three fixed placements executed against shared realisations. Left, the mobile event at $\sigma = 40$ s as the deadline tightens past the timing-certain backbone, where the ground-central and sky-proxy curves coincide; right, the emergency case against collapse severity. Thirty instances per point, 1200 realisations, 95% confidence intervals.

Table S3: Placement probe on the multi-hop continuum ($\sigma = 40$ s): delivery ratio by the deadline against the margin over the reliable path. Twenty instances, 1000 shared held-out realisations.

Margin	Ground	Proxy	Distributed	CGR + repair	SLA + repair
−60 s	0.633	0.580	0.000	0.661	0.743
0 s	0.748	0.662	0.000	0.825	0.876
60 s	0.908	0.623	0.000	0.875	0.924
120 s	0.974	0.786	0.000	0.889	0.976
240 s	0.994	0.555	0.000	0.905	0.998

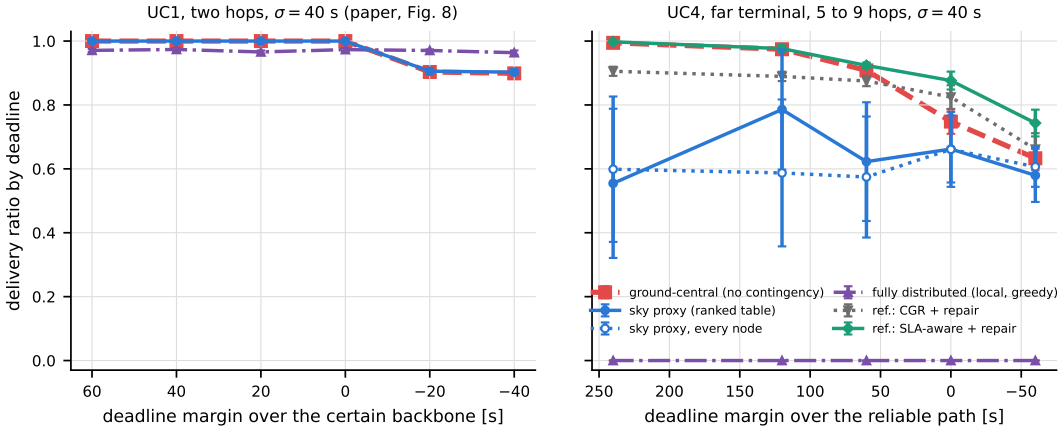


Figure S15: Placement probe on the mobile event (left) and on the multi-hop continuum (right) against the deadline margin, with CGR and the SLA-aware plan under hop-by-hop repair as references. "Sky proxy, every node" gives every node the proxy's ranked table.

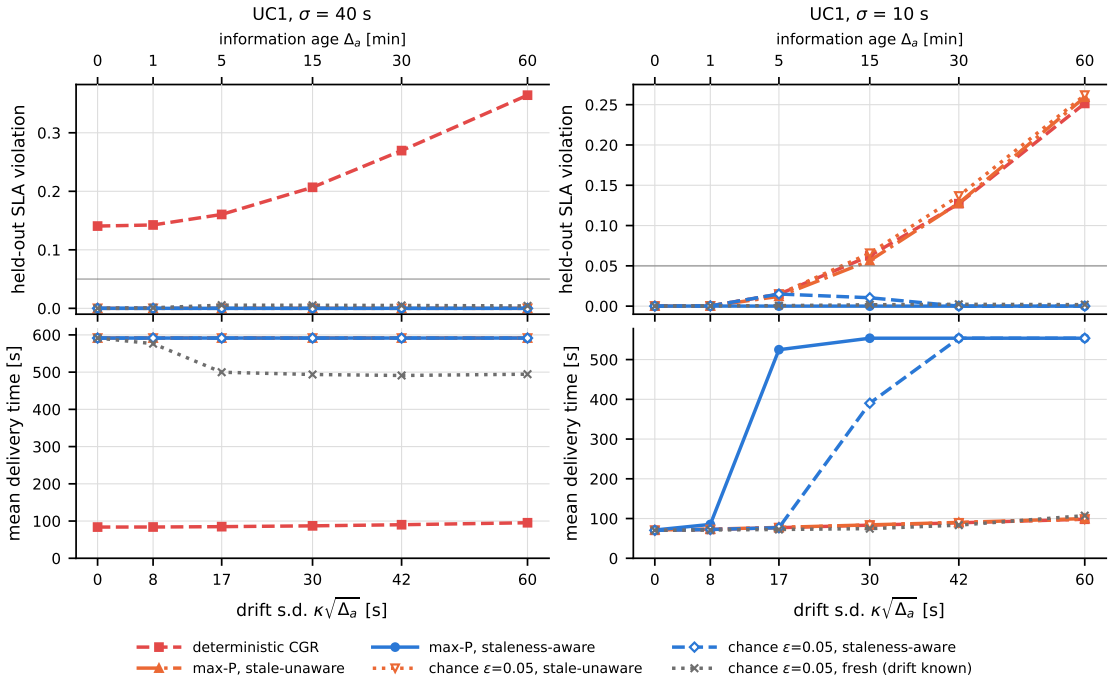


Figure S16: Planner information age on the mobile event at $\sigma = 40$ s (left) and 10 s (right): held-out violation (top) and mean delivery time (bottom) against the age Δ_a and the corresponding drift standard deviation $\kappa\sqrt{\Delta_a}$, for CGR and for SLA-aware planners (maximum probability and chance-constrained at $\varepsilon = 0.05$) that ignore or account for staleness, and a planner with fresh information.

Table S4: Make targets behind each result.

Target	Result
`run-uc1`, `run-uc2`, `run-uc3`, `run-uc4`	the four use cases (Fig. 5 of the paper, Fig. 7 of the paper, Fig. 8 of the paper, Fig. 9 of the paper)
`run-ablation`, `run-eps`	the corner ablation and the ε-sweep (Fig. 4 of the paper, Fig. 6 of the paper)
`run-joint_law`	the coupled contact (Fig. 2 of the paper)
`run-runtime`, `run-overhead`, `run-mc_scaling`	runtime, SABR comparison, forwarding cost (Fig. 10 of the paper)
`run-scenario`, `run-uc1_nonfair`	skeleton and fair-design check (Figs. S1, S2)
`run-compound`, `run-generality`	compounding and generality (Figs. S3, S4)
`run-gap`	candidate-set gap (Table S2, Fig. S5)
`run-rucop`	BRUF, RUCoP, probabilistic CGR (Figs. S6–S8)
`run-misspec_timing`, `run-misspec_capacity`, `run-dependence`, `run-estimation`	misspecified, correlated, and learned laws (Figs. S9–S12)
`run-flows`	concurrent bundles and storage (Fig. S13)
`run-placement`, `run-placement_uc4`	placement probe (Fig. S14, Table S3, Fig. S15)
`run-sidecar`, `run-staleness`	sidecar size and staleness (Fig. S16)